Formulae to use

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \Big[a_n \cos(nt) + b_n \sin(nt)\Big] \qquad (2\pi\text{-periodic})$$

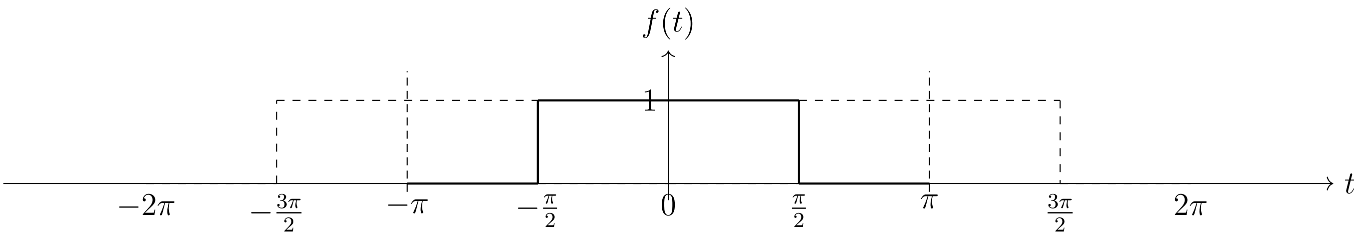
$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t)\, dt$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos(nt)\, dt, \qquad n \geq 1$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin(nt)\, dt, \qquad n \geq 1$$

Exercise 1 Find the Fourier series of the 2π-periodic rectangular pulse train.
Solution:

$$f(t) = \begin{cases} 1, & |t| < \frac{\pi}{2}, \\ 0, & \frac{\pi}{2} < |t| < \pi. \end{cases}$$



$$f(-t) = f(t) \qquad \Rightarrow \qquad b_n = 0$$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t)\, dt = \frac{2}{\pi} \int_{0}^{\pi} f(t)\, dt$$

$$= \frac{2}{\pi} \left(\int_0^{\pi/2} 1\, dt + \int_{\pi/2}^{\pi} 0\, dt \right) = \frac{2}{\pi} \cdot \frac{\pi}{2} = 1$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos(nt)\, dt = \frac{2}{\pi} \int_{0}^{\pi} f(t) \cos(nt)\, dt$$

$$= \frac{2}{\pi} \left(\int_0^{\pi/2} \cos(nt)\, dt + \int_{\pi/2}^{\pi} 0 \cdot \cos(nt)\, dt \right)$$

$$= \frac{2}{\pi} \int_0^{\pi/2} \cos(nt)\, dt = \frac{2}{\pi} \left. \frac{\sin(nt)}{n} \right|_0^{\pi/2}$$

$$= \frac{2}{\pi n} \sin\left(\frac{n\pi}{2}\right)$$

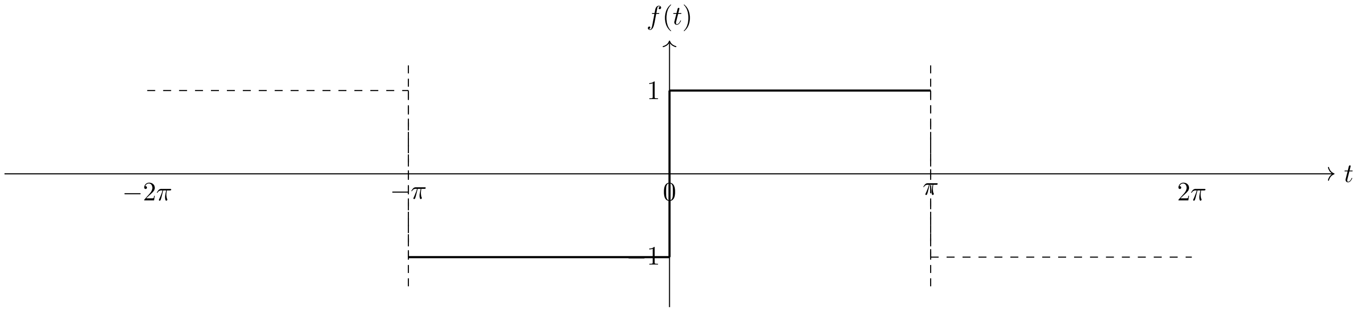
$$\boxed{a_n = \frac{2}{\pi n} \sin\left(\frac{n\pi}{2}\right), \qquad b_n = 0, \qquad a_0 = 1}$$

$$\sin\left(\frac{n\pi}{2}\right) = \begin{cases} 0, & n \text{ even}, \\ (-1)^k, & n = 2k+1 \end{cases}$$

$$\implies \quad \boxed{f(t) = \frac{1}{2} + \frac{2}{\pi}\sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1}\cos\big((2k+1)t\big)}$$

Alternative solution

$$f(t) = \begin{cases} -1, & -\pi < t < 0, \\ 1, & 0 < t < \pi. \end{cases}$$



$$f(-t) = -f(t) \qquad \Rightarrow \qquad a_0 = 0, \qquad a_n = 0$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\sin(nt)\,dt = \frac{2}{\pi}\int_0^{\pi} f(t)\sin(nt)\,dt$$

$$= \frac{2}{\pi}\int_0^{\pi} \sin(nt)\,dt$$

$$= \frac{2}{\pi}\; -\frac{\cos(nt)}{n}\bigg|_0^{\pi}$$

$$= \frac{2}{\pi n}\big(1 - \cos(n\pi)\big) = \frac{2}{\pi n}\big(1 - (-1)^n\big)$$

$$\implies \quad b_n = \begin{cases} 0, & n \text{ even}, \\ \dfrac{4}{\pi n}, & n \text{ odd} \end{cases}$$

$$\implies \quad \boxed{f(t) = \frac{4}{\pi}\sum_{k=0}^{\infty} \frac{1}{2k+1}\sin\big((2k+1)t\big)}$$
